

The ECS motherboard came with an accessory called Top-Hat flash. If for some reason the BIOS gets corrupted, all you need to do is place the Top-Hat flash tool on top of the BIOS chip, and you'll be able to boot and re-flash the BIOS and restore it. Similarly, Gigabyte, MSI and ASUS have a backup redundant BIOS facility on the motherboard so in case the BIOS gets corrupt, you may still be able to boot and restore the BIOS.

DFI has a unique feature known as CMOS Reloaded, in which you can set, name and save up to four different BIOS settings from within the CMOS setup. Then there is the Genie BIOS, which allows you to overclock to the max. Overclocking was standard on all the motherboards, where CPU core speed and memory voltage, along with HyperTransport Technology speed and DRAM speed, could be tweaked.

Performance

Gaming

The ECS KN1 Extreme, along with the DFI LanParty, were the best when it came to gaming performance. The other boards, though, were not far behind, and lagged by a maximum of five fps. The above two are the motherboards you might consider grabbing if you're a big-time gamer. The Gigabyte K8 Triton GA-K8N Pro-SLI posted disappointing scores here.

Synthetic benchmarks

In 3DMark05, there was no clear winner as such, because all the motherboards scored almost the same. This test is a general indicator of what kind of a gaming experience you'll get using a PC based on this motherboard. In this test, the Gigabyte K8 Triton again lagged way behind the others, especially in CPU marks.

In the PCMark 2004 test, the ASUS Proactive A8N-SLI and the Gigabyte 8Σ Series GA-K8NXP-SLI scored the highest—and almost the same—



Asus Proactive A8N-SLI

points, which reflects the fact that these motherboards will give overall better performance in daily use. The Gigabyte K8 Triton's woes continued: it failed to even complete this test after repeated attempts!

SiSoft Sandra 2005 saw the DFI and ECS motherboards score high in the memory benchmark, with the other boards not even coming close. However, in the CPU and file system benchmarks, almost all the boards scored equally.

Real-World Benchmarks

The ASUS A8N-SLI was the fastest in the video file encoding test at 110 seconds, followed by the ECS and the Gigabyte 8Σ Series at 111 seconds—practically no difference. These three boards are therefore better at utilising the power of the X2 to the maximum, so you can expect better performance in multimedia applications such as video editing programs.

The DFI LanParty UT NF4 SLI-D board performed better than the others in ZDBench Business Winstone 2004, scoring 31.1. Very close on its heels was the Gigabyte 8Σ Series

What Should You Look For In A High-End Motherboard?

Dual-core processor support: The motherboard must support dual-core processors of the highest speed available at the time of purchase, even if you plan to use a single-core processor. This is to ensure your computer will be future-proof—you will probably move on to a dual-core processor at some point.

High-speed memory support: The motherboard must support dual-channel memory of as high a speed as possible. Currently, motherboards support DDR memory of speeds as high as 400 MHz and DDR2 memory of speeds up to 888 MHz, such as the Gigabyte i-DNA GA-8I955X Royal.

Number of SATA ports: A higher number of SATA ports is preferred, especially since a SATA port supports one SATA device—unlike an IDE port, which can support two devices. Generally, four SATA ports are

found on most motherboards. Some may come with as many as 10.

SATA II support: The SATA standard, in which the data transfer rate goes up to 150 MBps, is slowly being replaced by SATA II, which boasts of 300 MBps. Motherboards that support SATA II also support the older SATA standard in case you have a SATA drive.

SLI: Even if you have or plan to buy a single PCI-Express graphics card, for future-proofing the graphics capabilities, a motherboard that supports SLI should be preferred. It provides the option to add another PCI-Express card to improve graphics performance in the future, in case a single card is not enough.

FireWire: A FireWire port is used to connect FireWire devices such as DV cams, usually to capture high-resolution digital video.

Dual Gigabit: Dual Gigabit ports allow you to be on two networks at the same time, so you can be connected to the Internet using one of the ports while the other port can be used to connect other computers on a home network or LAN.

Wi-Fi: Wi-Fi offers you the convenience of wireless networking and Internet access.

Bluetooth: Bluetooth is useful in situations such as when you wish to listen to music via your Bluetooth headset or when you want to transfer files between your PC and your Bluetooth mobile phone or PDA.

Overclocking features: A good premium-quality motherboard should come with advanced overclocking features. This is a feature much sought after by hardcore gamers, who would like to squeeze out all that the system is capable of.